Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/21/2006
PI Name:	Williams, Forman A. Ph.D.		
Project Title:	Scientific Support for a Proposed Space Shuttle Droplet Burning Experiment		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	858 534-5492
Organization Name:	University of California, San Diego		
PI Address 1:	Center for Energy Research		
PI Address 2:	Department of Mechanical and Aerospace Engineering		
PI Web Page:			
City:	La Jolla	**State:**	CA
Zip Code:	92093-0411	**Congressional District:**	53
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	06/08/2004	**End Date:**	12/31/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Vedha-Nayagam, Muthuswamy	**Contact Phone:**	216-433-8702
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Flight Program:	Shuttle/ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dryer, F.L. (Princeton University)		
Grant/Contract No.:	NNC04GB33G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This is a collaborative effort of NASA Glenn Research Center, the University of California, San Diego and Princeton University, to advance scientific knowledge of droplet combustion and to investigate flammability and suppression of combustion at reduced gravity. Experiments are planned for space platforms, including the Internationa l Space Station. By reducing buoyancy effects, space-based experiments contribute to improved understanding. In preparing for space experiments, ground-based microgravity experiments and theoretical and computational analyses are performed by the co-investigators, and relevant instrumentation is improved. The objective of this collaboration is to provide scientific support for droplet burning experiments to be performed on board space platforms. The specific objectives of these studies are to pro vide fundamental results which impact fire safety issues in low gravity environments and to advance the scientific understanding of droplet burning/flame/extinction dynamics over characteristic times which can not be studied in ground based configurations. Engineering and design support functions contribute to: developing and refining methods to perform isolated droplet combustion experiments in general; determining and testing methods to acquire		

	appropriate experimental observations; developing and evaluating methods for presenting and analyzing the collected data. Experimental support functions include: theoretical analyses to define appropriate experimental conditions and to analyze the experimental results obtained; bench scale and drop tower experiments to test and verify methodologies to be applied in the space-based experiments. The overall purpose of the research is to achieve fundamental advances that will aid in improving fire safety in space environments. In particular, unsteady liquid and gas phenomenon, and extinction chemistry of normal alkanes and simple alcohols are investigated.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	A scientific review was presented successfully. A new test matrix was completed, involving studies of droplet combustion in different atmospheres, directed towards evaluating fire-safety hazards and mechanisms of fire suppression. In particular, because of the fire-suppression systems on ISS, atmospheres containing carbon dioxide are to be studied. The results are intended to improve knowledge of combustion relevant to fire protection in space environments.
Bibliography Type:	Description: (Last Updated: 03/07/2024)